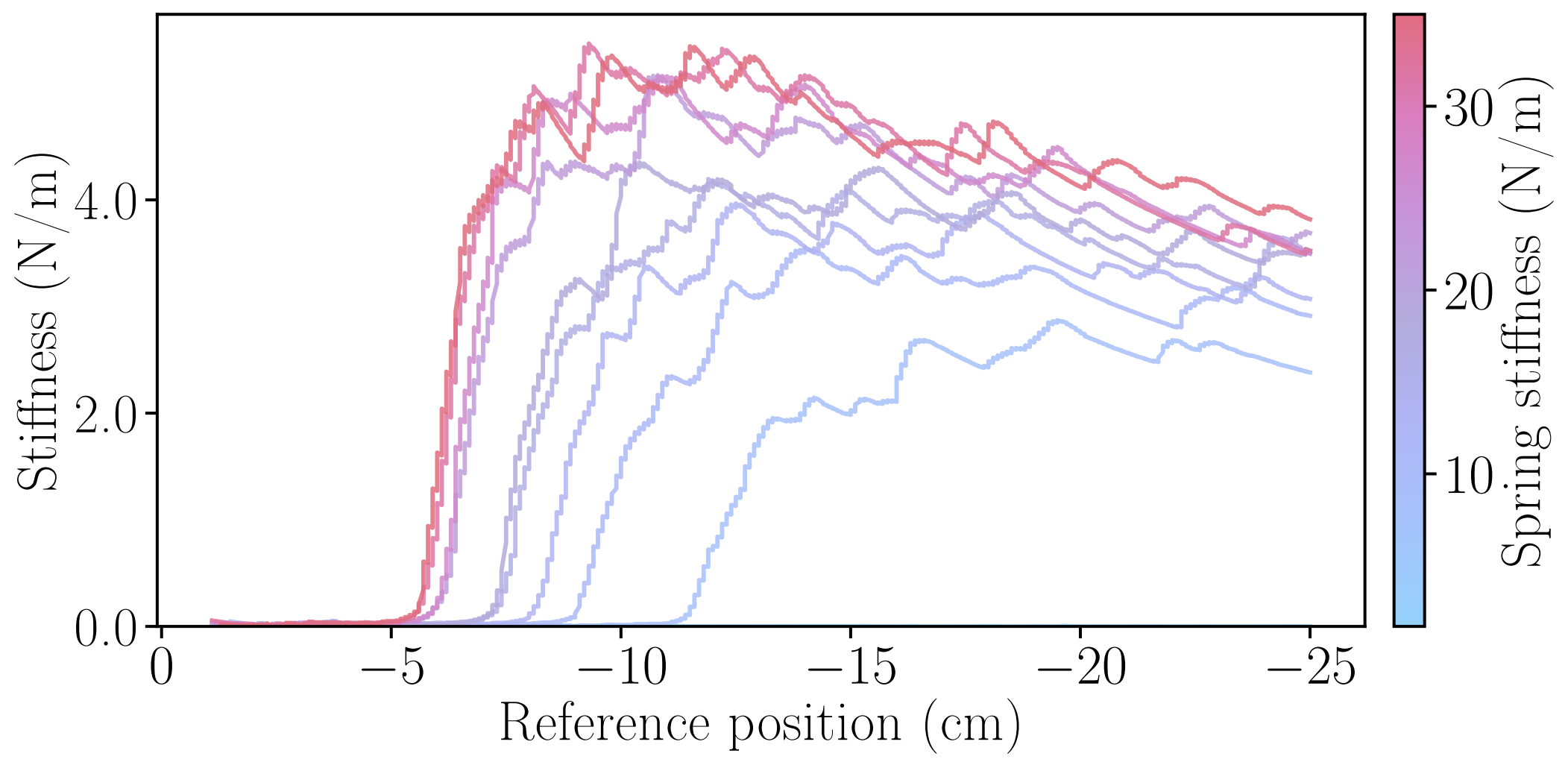

Stiffness (N/m)
0.0
2.0
4.0
0
−5
−10
−15
−20
−25
Reference position (cm)
Spring stiffness (N/m)
10
20
30